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SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 30%

QUESTIONS: 5

Annexure A

Total Marks: 25

Comparative Analysis

1. Star topology is generally more expensive than bus topology because it requires a central device such as a switch or hub and more cabling to connect each node individually. In contrast, bus topology uses a single backbone cable, making it low-cost and easier to install. However, in terms of reliability, star topology is superior because failure of one node does not affect others, whereas a failure in the main bus cable can bring down the entire network. Performance is also better in star topology since data transmission is managed through a central device, reducing collisions, while bus topology suffers from performance degradation as more devices are added due to increased data traffic on the shared medium.
2. Mesh topology offers very high fault tolerance because every node is connected to multiple other nodes, providing multiple paths for data transmission. If one link fails, data can still be routed through alternate paths without disrupting communication. In contrast, star topology has moderate fault tolerance; while failure of individual nodes does not affect the network, failure of the central hub or switch can cause the entire network to stop functioning. Therefore, mesh topology is more robust and reliable in critical environments compared to star topology.
3. Bus topology is simple and easy to install because it uses a single backbone cable with minimal connections, making it suitable for small networks. On the other hand, mesh topology is highly complex to install as each node must be connected to every other node, requiring a large number of cables and ports. This complexity increases significantly as the number of devices grows, making mesh topology time-consuming and costly to implement compared to the straightforward setup of bus topology.
4. Hybrid topology is highly scalable because it combines two or more different topologies (such as star, mesh, or bus), allowing the network to expand flexibly according to organizational needs. New segments can be added without affecting the existing network structure. In comparison, star topology is also scalable but to a

Comparative Analysis

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ANSWER:

Star topology is generally more expensive than bus topology because it requires a **central switch or hub** and separate cables for connecting every device. Bus topology uses a single backbone cable, so it requires less cabling and fewer networking devices, making its initial installation cheaper.

However, star topology provides better **reliability and performance**. If the cable connecting one computer fails, only that particular computer is disconnected, while the rest of the network continues operating normally. In bus topology, a failure in the main backbone cable can interrupt communication for all connected devices.

Star topology also provides better performance because the switch manages communication between devices and reduces unnecessary traffic. In a bus network, all devices share the same communication medium, so collisions and network congestion can increase as more devices are added.

Comparison:

Feature	Star	Bus
Cost	Moderate/High	Low
Reliability	High	Low
Performance	High	Moderate/Low
Fault Isolation	Easy	Difficult
Expansion	Easy	Difficult
Maintenance	Easy	Difficult

Conclusion: Star topology is more suitable for modern networks where reliability and performance are more important than minimum installation cost.

2. Mesh topology offers very high fault tolerance because every node is connected to multiple other nodes, providing multiple paths for data transmission. If one link fails, data can still be routed through alternate paths without disrupting communication. In contrast, star topology has moderate fault tolerance; while failure of individual nodes does not affect the network, failure of the central hub or switch can cause the entire network to stop functioning. Therefore, mesh topology is more robust and reliable in critical environments compared to star topology.

ANSWER:

Mesh topology provides **very high fault tolerance** because nodes have multiple connections and alternative paths are available for data transmission. If one link fails, network traffic can use another available path.

Star topology has moderate fault tolerance. Failure of a single cable or end device affects only that device, which is an important advantage. However, the central switch is a critical point of failure. If the central switch fails, all devices connected through it may lose network connectivity.

Mesh topology is therefore preferred for critical environments such as **banking, data centers, military networks, and telecommunications**, where continuous operation is important.

The major disadvantage of mesh topology is its high cost and complexity because it requires many physical or logical connections.

Comparison:

Feature	Mesh	Star
Fault Tolerance	Very High	Moderate
Redundancy	Very High	Low/Moderate
Cost	Very High	Moderate
Complexity	High	Low
Troubleshooting	Difficult	Easy
Critical Applications	Excellent	Good

Conclusion: Mesh topology is more robust than star topology, while star topology provides a better balance between reliability, cost, and simplicity.

3.
- Bus topology is simple and easy to install because it uses a single backbone cable with minimal connections, making it suitable for small networks. On the other hand, mesh topology is highly complex to install as each node must be connected to every other node, requiring a large number of cables and ports. This complexity increases significantly as the number of devices grows, making mesh topology time-consuming and costly to implement compared to the straightforward setup of bus topology.

ANSWER:

Bus topology is simple and inexpensive to install because multiple devices share a **common backbone cable**. It requires fewer cables and networking components, making it suitable for small and temporary networks.

Mesh topology is significantly more complex because devices require multiple connections to other devices. In a full mesh network, the number of links increases rapidly as more nodes are added.

For example, a network containing only a few devices can already require many connections in a full mesh. This results in higher cable requirements, more ports, higher installation costs, and more complicated configuration.

Bus topology is therefore easier to deploy, while mesh topology is preferred when redundancy and fault tolerance are more important than installation simplicity.

Comparison:

Feature	Bus	Mesh
Installation	Easy	Complex
Cabling	Low	Very High
Cost	Low	Very High
Configuration	Simple	Complex
Scalability	Poor	Good
Redundancy	Very Low	Very High

Conclusion: Bus topology is easier and cheaper to install, whereas mesh topology provides superior redundancy at the cost of greater complexity.

4. Hybrid topology is highly scalable because it combines two or more different topologies (such as star, mesh, or bus), allowing the network to expand flexibly according to organizational needs. New segments can be added without affecting the existing network structure. In comparison, star topology is also scalable but to a limited extent, as it depends on the capacity of the central device (switch or hub). Once the central device reaches its port limit, expansion requires additional hardware. Thus, hybrid topology provides greater scalability than star topology.

ANSWER:

Hybrid topology combines **two or more different network topologies**, such as star, mesh, ring, or bus. This allows organizations to design different parts of the network according to their specific requirements.

Hybrid topology is highly scalable because new network segments can be added without completely redesigning the existing infrastructure. For example, a university can use a mesh backbone, star topology within buildings, and different structures inside individual laboratories.

Star topology is also relatively easy to expand, but its scalability depends on the number of available ports on the central switch. When all switch ports are occupied, an additional switch may be required.

Hybrid networks provide greater flexibility for large organizations, but they require more planning and skilled administration.

Comparison:

Feature	Hybrid	Star
Scalability	Very High	High
Flexibility	Very High	Moderate
Cost	Moderate/High	Moderate
Design Complexity	High	Low
Expansion	Very Flexible	Switch-dependent
Suitable For	Large Networks	Small/Medium Networks

Conclusion: Hybrid topology provides greater scalability and flexibility, making it suitable for large organizations and campus networks.

5. In terms of maintenance, star topology is relatively easy to manage because issues can be quickly identified and isolated to a specific node or cable. Mesh topology, while highly reliable, is difficult to maintain due to its complex structure and large number of connections, making troubleshooting time-consuming. Bus topology is harder to maintain than star because faults in the backbone cable can disrupt the entire network and are difficult to locate. Hybrid topology has moderate maintenance complexity; while it benefits from flexibility, managing multiple combined topologies requires careful monitoring and skilled administration.

ANSWER:

Star topology is relatively easy to **maintain and troubleshoot** because every device has an individual connection to the central switch. When a problem occurs, administrators can check the specific cable, port, or device without affecting the rest of the network.

Mesh topology is highly reliable but difficult to maintain because it contains many connections. Identifying a faulty link or configuration problem can take considerable time, particularly in large networks.

Bus topology is also difficult to troubleshoot because a problem in the backbone cable can affect multiple devices. Locating the exact point of failure may require checking the cable and connections across the entire network.

Hybrid topology has moderate to high maintenance requirements because different sections may use different technologies and topologies. Network administrators must understand and monitor each individual section.

Comparison:

Feature	Star	Mesh	Bus	Hybrid
Maintenance	Easy	Difficult	Difficult	Moderate
Troubleshooting	Easy	Complex	Difficult	Moderate
Reliability	High	Very High	Low	High
Management	Simple	Complex	Simple/Moderate	Complex
Scalability	High	Moderate	Low	Very High

Conclusion: Star topology is generally the easiest to maintain, while mesh provides the highest redundancy. Hybrid topology offers flexibility and scalability but requires more skilled network administration.

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Concepts	15%	Demonstrates complete and in-depth understanding of all concepts being compared	Good understanding with minor gaps	Basic understanding; some concepts unclear	Poor understanding of concepts
Comparison Depth	20%	Provides detailed, insightful, and meaningful comparisons across multiple aspects	Adequate comparison with relevant points	Limited comparison; lacks depth	Minimal or incorrect comparison
Criteria Selection	10%	Selects highly relevant and appropriate comparison parameters	Mostly relevant criteria	Some irrelevant or missing criteria	Poor or no criteria selection
Analytical Skills	15%	Strong analysis with logical reasoning and clear justification	Good analysis with some justification	Basic analysis with weak reasoning	No clear analysis
Organization & Structure	10%	Well-organized, clear, and logically structured	Generally organized	Somewhat disorganized	Poorly structured
Use of	10%	Uses strong,	Uses some	Limited	No supporting

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Examples / Evidence		relevant examples or data to support comparison	examples	examples	examples

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Clarity of Presentation	10%	Very clear, concise, and easy to understand	Mostly clear	Somewhat unclear	Difficult to understand
Conclusion & Insights	10%	Insightful conclusion with strong justification	Clear conclusion	Basic conclusion	No or weak conclusion

	A	B	C	D	E	F	G	H	Total
Q ₁	1.5	2	1	1.5	1	1	0.5	1	9.5
Q ₂	1	2	1	1	1	0.5	1	0.5	8
Q ₃	1.5	1.5	0.5	1.5	1	1	0.5	1	8.5
Q ₄	1	1	0.5	1	0.5	1	0.5	0.5	6
Q ₅	1.5	2	1	1.5	1	0.5	1	1	9.5

41.5